HARDWARE EXPERIMENT – 20

❖ Experimental Investigation of Microwave Wireless-Link Performance and Electromagnetic Interference Effects Using an RF Communication Test Setup

➔ CO Mapping: CO5

Bloom's Level: K5 – Evaluate

WK Level: WK6 – RF link and EMI/EMC system evaluation

➔ Objectives of this Experiment:

1. To establish a microwave/RF wireless communication link and measure received-signal characteristics.
2. To investigate the effect of distance, antenna orientation and environmental conditions on link performance.
3. To introduce controlled interference or unwanted RF signals and observe their effect on communication quality.
4. To analyze the results in the context of electromagnetic interference and electromagnetic compatibility requirements.

➔ Expected Outcome from this Experiment:

Students shall be able to:

1. Establish and operate a microwave/RF wireless communication link and measure the corresponding received-signal characteristics.
2. Measure and analyze the effect of distance, antenna orientation and environmental conditions on wireless link performance.
3. Investigate and evaluate the effect of controlled RF interference on communication quality and identify the resulting performance degradation.
4. Analyze and interpret the experimental results in the context of electromagnetic interference (EMI) and electromagnetic compatibility (EMC) requirements.

➔ Hardware Required:

RF/microwave signal source, transmitting and receiving antennas, spectrum analyzer or SDR,

power meter/receiver and RF accessories.

ECA13 Microwave Engineering

➔ Experimental Procedure:

A. Wireless-Link Characterization: Establish a 2.4 GHz RF link with Pt = 10 dBm and measure

received power and SNR at d = 1, 2, 3 and 5 m for antenna orientations 0°, 45° and 90°. Tabulate all

readings, identify the best and worst link condition, and explain the effect of distance and antenna

orientation on received signal strength.

B. Controlled EMI Analysis: Keep the 2.4 GHz link at d = 3 m and Pt = 10 dBm. First record the

baseline received power and SNR; then apply a faculty-approved controlled interference signal of Pi =

−20 dBm near the operating channel. Record the new received power/SNR, calculate ΔSNR and ΔPr,

and explain the observed degradation using EMI/EMC principles.

```
clc;
clear;
close all;

%% EXPERIMENT 20
% Wireless-Link Performance and EMI Analysis
% ECA13 Microwave Engineering

%% 1. INPUT PARAMETERS FROM LAB MANUAL

f = 2.4e9;          % Frequency (Hz)
Pt = 10;            % Transmit power (dBm)
Pi = -20;           % Interference power (dBm)
```

```
c = 3e8;
lambda = c/f;

distances = [1 2 3 5];
angles = [0 45 90];

noiseFloor = -90;         % Assumed receiver noise floor (dBm)
crossIsolation = 20;      % Assumed cross-polarization isolation (dB)

%% 2. ANTENNA ORIENTATION MODEL

% Relative polarization coupling
orientationLoss = -10*log10( ...
    cosd(angles).^2 + ...
    10^(-crossIsolation/10)*sind(angles).^2);

%% 3. FREE SPACE PATH LOSS

FSPL = zeros(1,length(distances));

for i = 1:length(distances)

    FSPL(i) = 20*log10(4*pi*distances(i)/lambda);

end

%% 4. WIRELESS LINK CHARACTERIZATION
```

```
Pr = zeros(length(distances),length(angles));
SNR = zeros(length(distances),length(angles));

for i = 1:length(distances)

    for j = 1:length(angles)

        % Received signal power
        Pr(i,j) = Pt - FSPL(i) + orientationLoss(j);

        % Signal-to-noise ratio
        SNR(i,j) = Pr(i,j) - noiseFloor;

    end

end

%% 5. BEST AND WORST LINK CONDITIONS

[maxPower,idxBest] = max(Pr(:));
[minPower,idxWorst] = min(Pr(:));

[bestDistance,bestAngle] = ind2sub(size(Pr),idxBest);
[worstDistance,worstAngle] = ind2sub(size(Pr),idxWorst);

bestSNR = SNR(bestDistance,bestAngle);
worstSNR = SNR(worstDistance,worstAngle);

%% 6. CONTROLLED EMI ANALYSIS AT 3 m, 0 DEG
```

```
dEMI = 3;
thetaEMI = 0;

% Free-space path loss
FSPL_EMI = 20*log10(4*pi*dEMI/lambda);

% Baseline carrier power
PrBaseline = Pt - FSPL_EMI;

% Baseline SNR
SNRBaseline = PrBaseline - noiseFloor;

% Received interference power
PrInterference = Pi - FSPL_EMI;

% Convert dBm to mW
noise_mW = 10^(noiseFloor/10);
signal_mW = 10^(PrBaseline/10);
interference_mW = 10^(PrInterference/10);

% Total noise and interference
totalNoise_mW = noise_mW + interference_mW;

% SNR with interference
SNRInterference = 10*log10(signal_mW/totalNoise_mW);

% SNR degradation
deltaSNR = SNRBaseline - SNRInterference;
```

```
% Carrier power is unchanged in this ideal model
deltaPrCarrier = PrBaseline - PrBaseline;

% Composite in-channel power
PbaselineTotal = 10*log10(signal_mW + noise_mW);

PinterferenceTotal = 10*log10( ...
    signal_mW + interference_mW + noise_mW);

% Composite in-channel power change
deltaPrComposite = PbaselineTotal - PinterferenceTotal;

% SNR degradation percentage
degradationPercent = ...
    deltaSNR/SNRBaseline*100;

%% 7. SINGLE FIGURE WITH ALL OUTPUTS

figure('Color','w',...
       'Name','ECA13 Experiment 20',...
       'NumberTitle','off');

tiledlayout(3,2,'TileSpacing','compact',...
            'Padding','compact');

sgtitle('Wireless-Link Performance and EMI Analysis');

%% GRAPH 1: RECEIVED POWER
```

```
nexttile;

bar(distances,Pr,'grouped');

xlabel('Distance (m)');
ylabel('Received Power (dBm)');

title('Received Power vs Distance');

legend('0 deg','45 deg','90 deg',...
       'Location','best');

grid on;

%% GRAPH 2: SNR

nexttile;

plot(distances,SNR(:,1),'bo-',...
    'LineWidth',1.5);

hold on;

plot(distances,SNR(:,2),'rs-',...
    'LineWidth',1.5);

plot(distances,SNR(:,3),'k^-',...
    'LineWidth',1.5);
```

```
xlabel('Distance (m)');
ylabel('SNR (dB)');

title('SNR vs Distance');

legend('0 deg','45 deg','90 deg',...
       'Location','best');

grid on;

%% GRAPH 3: ANTENNA ORIENTATION

nexttile;

bar(angles,Pr(3,:));

xlabel('Antenna Orientation (degrees)');
ylabel('Received Power (dBm)');

title('Orientation Effect at 3 m');

set(gca,'XTick',angles);

grid on;

%% GRAPH 4: FREE SPACE PATH LOSS

nexttile;
```

```
plot(distances,FSPL,'mo-',...
     'LineWidth',2);

xlabel('Distance (m)');
ylabel('Path Loss (dB)');

title('Free-Space Path Loss');

grid on;

%% GRAPH 5: EMI COMPARISON

nexttile;

bar([SNRBaseline SNRInterference]);

set(gca,'XTick',1:2,...
    'XTickLabel',{'Baseline','With EMI'});

ylabel('SNR (dB)');

title('Controlled EMI Effect');

grid on;

%% GRAPH 6: EXPERIMENTAL RESULTS

nexttile;
```

```
axis off;

text(0.02,0.96,'EXPERIMENT 20 RESULTS',...
    'FontSize',11,'FontWeight','bold');

text(0.02,0.83,...
    sprintf('Best link: %.0f m, %.0f deg',...
    distances(bestDistance),angles(bestAngle)));

text(0.02,0.70,...
    sprintf('Best Pr: %.2f dBm',maxPower));

text(0.02,0.57,...
    sprintf('Best SNR: %.2f dB',bestSNR));

text(0.02,0.44,...
    sprintf('Worst link: %.0f m, %.0f deg',...
    distances(worstDistance),angles(worstAngle)));

text(0.02,0.31,...
    sprintf('Worst Pr: %.2f dBm',minPower));

text(0.02,0.18,...
    sprintf('Baseline SNR: %.2f dB',SNRBaseline));

text(0.02,0.05,...
    sprintf('EMI SNR: %.2f dB',SNRInterference));
```

```
%% 8. COMMAND WINDOW RESULTS

disp('========================================');
disp('WIRELESS LINK CHARACTERIZATION');
disp('========================================');

disp('Received Power (dBm):');

disp(array2table(Pr,...
    'VariableNames',{'Orientation_0',...
                     'Orientation_45',...
                     'Orientation_90'},...
    'RowNames',{'Distance_1m',...
                'Distance_2m',...
                'Distance_3m',...
                'Distance_5m'}));

disp('SNR (dB):');

disp(array2table(SNR,...
    'VariableNames',{'Orientation_0',...
                     'Orientation_45',...
                     'Orientation_90'},...
    'RowNames',{'Distance_1m',...
                'Distance_2m',...
                'Distance_3m',...
                'Distance_5m'}));

fprintf('\nBEST LINK\n');
```

```
fprintf('Distance: %.1f m\n',distances(bestDistance));
fprintf('Orientation: %.0f degrees\n',angles(bestAngle));
fprintf('Received Power: %.2f dBm\n',maxPower);
fprintf('SNR: %.2f dB\n',bestSNR);

fprintf('\nWORST LINK\n');
fprintf('Distance: %.1f m\n',distances(worstDistance));
fprintf('Orientation: %.0f degrees\n',angles(worstAngle));
fprintf('Received Power: %.2f dBm\n',minPower);
fprintf('SNR: %.2f dB\n',worstSNR);

%% EMI RESULTS

fprintf('\nCONTROLLED EMI ANALYSIS AT 3 m\n');

fprintf('Transmit Power: %.2f dBm\n',Pt);
fprintf('Interference Power: %.2f dBm\n',Pi);

fprintf('Baseline Received Power: %.2f dBm\n',...
    PrBaseline);

fprintf('Received Interference Power: %.2f dBm\n',...
    PrInterference);

fprintf('Baseline SNR: %.2f dB\n',...
    SNRBaseline);

fprintf('Interference SNR: %.2f dB\n',...
    SNRInterference);
```

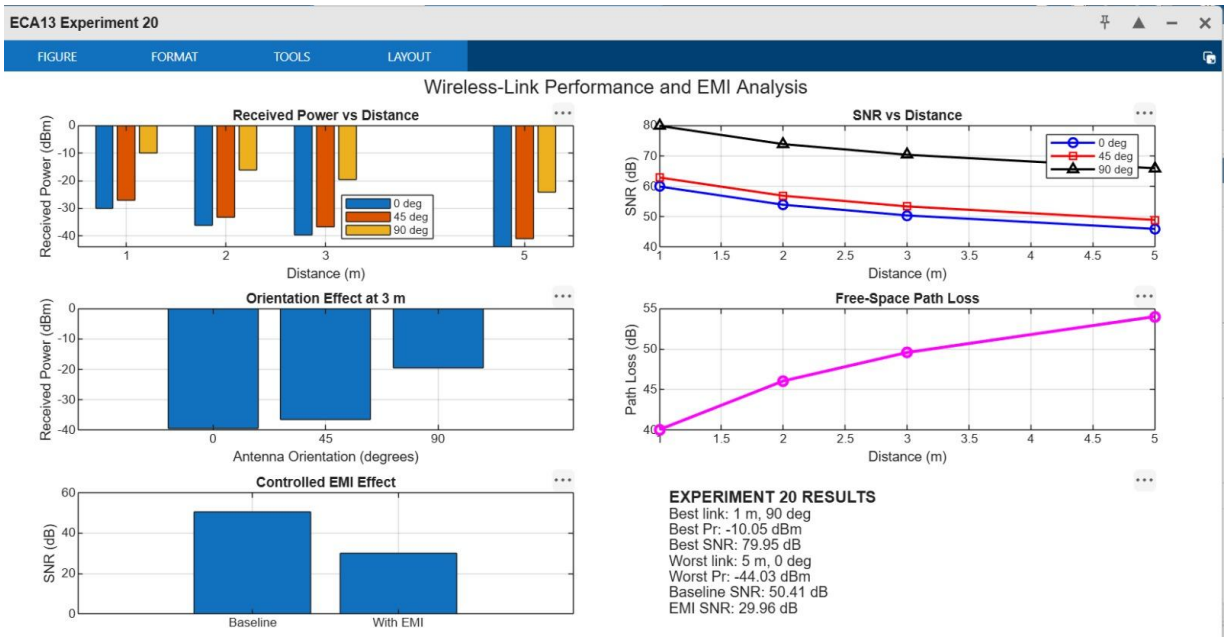
```
fprintf('Delta SNR: %.2f dB\n',...
    deltaSNR);

fprintf('Delta Carrier Power: %.2f dB\n',...
    deltaPrCarrier);

fprintf('Delta Composite Power: %.2f dB\n',...
    deltaPrComposite);

fprintf('SNR Degradation: %.2f %%\n',...
    degradationPercent);

disp('SIMULATION COMPLETED');
```




```
==========================================
WIRELESS LINK CHARACTERIZATION
==========================================
Received Power (dBm):
                Orientation_0    Orientation_45    Orientation_90
```

Distance_1m	-30.046	-27.079	-10.046
Distance_2m	-36.067	-33.1	-16.067
Distance_3m	-39.588	-36.621	-19.588
Distance_5m	-44.025	-41.058	-24.025

SNR (dB):

	Orientation_0	**Orientation_45**	**Orientation_90**
Distance_1m	59.954	62.921	79.954
Distance_2m	53.933	56.9	73.933
Distance_3m	50.412	53.379	70.412
Distance_5m	45.975	48.942	65.975

BEST LINK
Distance: 1.0 m
Orientation: 90 degrees
Received Power: -10.05 dBm
SNR: 79.95 dB

WORST LINK
Distance: 5.0 m
Orientation: 0 degrees
Received Power: -44.03 dBm
SNR: 45.97 dB

CONTROLLED EMI ANALYSIS AT 3 m
Transmit Power: 10.00 dBm
Interference Power: -20.00 dBm
Baseline Received Power: -39.59 dBm
Received Interference Power: -69.59 dBm
Baseline SNR: 50.41 dB
Interference SNR: 29.96 dB
Delta SNR: 20.45 dB
Delta Carrier Power: 0.00 dB
Delta Composite Power: -0.00 dB
SNR Degradation: 40.57 %
SIMULATION COMPLETED